Ambiguity Theory

A Mathematical Framework for
Structural Indeterminacy

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Abstract

We introduce *Ambiguity Theory*, a mathematical framework for handling structures that are *productively indeterminate*: not probabilistically uncertain, not logically contradictory, but structurally ambiguous in a sense that carries genuine information. We define *ambi-objects* as the primitive entities of the theory, develop their algebra, and characterise the corresponding category **Amb**. Our central result connects Ambiguity Theory rigorously to Bayesian probability: an ambi-object is precisely the pre-image fibre of a Bayesian update map at a point in the interior of the prior simplex. This adjoint relationship means every theorem in Bayesian inference has a dual theorem in Ambiguity Theory. We further develop coupling structure, ambiguity profiles, a resolution-complexity measure, and algorithms for ambi-object manipulation. Applications to natural-language parsing, quantum-state interpretation, and machine-learning hypothesis spaces are discussed.

The treatise ends with "The End"

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1 Introduction

Classical mathematics offers several tools for modelling incompleteness of knowledge. Probability theory assigns weights to exhaustive alternatives; fuzzy logic [15] assigns membership degrees; non-deterministic automata admit multiple transitions. In each case, indeterminacy is either *quantified* (probability) or *extended* (fuzzy membership), but ultimately the goal remains resolution to a definite value.

We argue that a third phenomenology exists: *structural ambiguity*, where the multiplicity of interpretations is not a deficiency to be corrected but the most complete description available. The canonical illustration is the English sentence

"I saw the man with the telescope."

This expression is simultaneously two parse trees, carrying strictly more information than either parse alone. Assigning a probability of 0.5 to each destroys the structural fact that *both readings are fully grammatical and simultaneously active in the utterance.*

Our framework, *Ambiguity Theory*, treats this irreducible multiplicity as a first-class mathematical object. Section 2 recalls relevant background. Section 3 defines ambi-objects axiomatically. Section 4 develops their algebra and category. Section 5 establishes the Bayesian inversion correspondence. Section 6 treats coupling and independence. Section 7 extends to continuous ambiguity and sheaf cohomology. Section 9 provides algorithms. Section 10 discusses applications in data science and machine learning. Section 13 states open problems.

2 Preliminaries

2.1 Probability Simplex

Let $[n] = \{1, \ldots, n\}$. The $(n-1)$-*simplex* is

$$\Delta^{n-1} \;=\; \left\{ p \in \mathbb{R}^n_{\geq 0} \;\middle|\; \textstyle\sum_{i=1}^n p_i = 1 \right\}.$$

Its relative interior $\Delta^{n-1\circ}$ consists of strictly positive probability vectors. The *vertices* e_i represent certainty in hypothesis i. We follow the subjective probability framework of de Finetti [4].

2.2 Bayesian Update

Let $H = \{h_1, \ldots, h_n\}$ be a finite hypothesis space and E an evidence variable with likelihood function $\mathcal{L}(E{=}e \mid h_i) = \ell_i > 0$ for all i. The *Bayesian update map* is

$$\mathcal{B}_e : \Delta^{n-1} \longrightarrow \Delta^{n-1}, \qquad (\mathcal{B}_e\pi)_i \;=\; \frac{\ell_i\,\pi_i}{\displaystyle\sum_{j=1}^n \ell_j\,\pi_j}.$$

$\mathcal{B}_e$ is a smooth map; its Jacobian encodes the local sensitivity of posterior beliefs to prior perturbations [7].

2.3 Sheaf-Theoretic Background

Let X be a topological space. A *sheaf* $\mathcal{S}$ on X assigns to each open set $U \subseteq X$ a set $\mathcal{S}(U)$ of *sections*, with restriction maps satisfying the gluing axiom. The *first Čech cohomology group* $H^1(X, \mathcal{S})$ measures the obstruction to assembling locally consistent sections into a global section [6]. We exploit this in Section 7.

2.4 Category Theory

We assume familiarity with categories, functors, and natural transformations at the level of MacLane [10]; the foundational language of natural equivalences goes back to Eilenberg & MacLane [5] and the functorial semantics perspective to Lawvere [9]. We recall that a *topos* is a category that behaves like the category of sets: it has all finite limits, exponentials, and a subobject classifier [8].

3 Ambi-Objects

3.1 Definition and Axioms

Definition 3.1 (Ambi-object)**.** An *ambi-object* over a base set $\mathcal{U}$ is a formal expression

$$A \;=\; \langle a_1 \mid a_2 \mid \cdots \mid a_n\rangle, \qquad a_i \in \mathcal{U}, \quad n \geq 1,$$

where the a_i are called the *branches* of A. Every $u \in \mathcal{U}$ embeds as the *definite* ambi-object $\langle u\rangle$.

The theory is governed by five axioms.

Axiom 1 (Non-collapsibility)**.** No internal derivation rule selects a branch: there is no proof $\langle a_1 \mid \cdots \mid a_n\rangle = a_i$ within the theory for $n \geq 2$.

Axiom 2 (Branch symmetry)**.** Branches are unordered: $\langle a_1 \mid \cdots \mid a_n\rangle \cong \langle a_{\sigma(1)} \mid \cdots \mid a_{\sigma(n)}\rangle$ for every permutation $\sigma \in S_n$.

Axiom 3 (Non-flattenability)**.** Nesting is not mere aggregation:

$$\langle\langle a \mid b\rangle \mid c\rangle \;\neq\; \langle a \mid b \mid c\rangle.$$

The left-hand side is ambiguous between an ambiguous object and c; the right-hand side is three-way ambiguous.

Axiom 4 (Functoriality)**.** A map $f : \mathcal{U} \to \mathcal{V}$ is *ambiguity-preserving* if

$$f\big(\langle a_1 \mid \cdots \mid a_n\rangle\big) \;=\; \langle f(a_1) \mid \cdots \mid f(a_n)\rangle.$$

Maps that do not satisfy this identity are called *resolving*.

Axiom 5 (Resolution)**.** There exists an external family of operators $\mathcal{R}_i$, $1 \leq i \leq n$, such that $\mathcal{R}_i\big(\langle a_1 \mid \cdots \mid a_n\rangle\big) = a_i$. Resolution is well-defined but *lossy*: the original ambi-object cannot be recovered from a_i alone.

3.2 Ambiguity Profile

Definition 3.2 (Depth and Width)**.** Define $\mathrm{depth}(u) = 0$ for $u \in \mathcal{U}$ and

$$\mathrm{depth}\big(\langle a_1 \mid \cdots \mid a_n\rangle\big) \;=\; 1 + \max_{1\leq i\leq n} \mathrm{depth}(a_i).$$

The *width* at level k is the number of branches at depth k. The *ambiguity profile* $\mathrm{profile}(A)$ is the ordered sequence $\big(\mathrm{width}_0(A), \mathrm{width}_1(A), \ldots, \mathrm{width}_{\mathrm{depth}(A)}(A)\big)$, equivalently represented as a rooted ordered tree.

Definition 3.3 (Ambiguity Isomorphism)**.** Two ambi-objects A and B are *ambiguity-isomorphic*, written $A \simeq_{\mathrm{amb}} B$, if $\mathrm{profile}(A) = \mathrm{profile}(B)$ as labelled trees.

Example 3.4. Let $A = \langle\langle p \mid q\rangle \mid r\rangle$ and $B = \langle\langle x \mid y\rangle \mid z\rangle$. Then $\mathrm{profile}(A) = \mathrm{profile}(B)$ (both are a depth-2 tree with root branching factor 2 and one internal node of branching factor 2), so $A \simeq_{\mathrm{amb}} B$, even though their content differs.

4 Algebra and Category of Ambi-Objects

4.1 Operations on Ambi-Objects

Definition 4.1 (Merge and Split)**.** The *merge* of ambi-objects $A = \langle a_1 \mid \cdots \mid a_m \rangle$ and $B = \langle b_1 \mid \cdots \mid b_n \rangle$ is

$$A \oplus B \;=\; \langle a_1 \mid \cdots \mid a_m \mid b_1 \mid \cdots \mid b_n \rangle.$$

Note $A \oplus B = B \oplus A$ (by Axiom 2). The *nest* of A inside B at branch j is obtained by replacing b_j with A, yielding a new ambi-object of greater depth.

Proposition 4.2. $(\mathbf{AObj}, \oplus, \langle \cdot \rangle)$ *is a commutative monoid, where* $\langle \cdot \rangle$ *denotes the singleton (definite) ambi-object serving as identity.*

Proof. Commutativity follows from Axiom 2. Associativity holds by inspection since branch lists concatenate associatively. The identity $\langle u \rangle \oplus A = A$ follows from identifying the singleton branch with the ambient element u and checking that no new branching structure is introduced when $n = 1$. □

4.2 The Category Amb

Definition 4.3 (Category of Ambi-Objects)**.** The category **Amb** has:

- **Objects:** all ambi-objects over $\mathcal{U}$.
- **Morphisms:** ambiguity-preserving maps (Definition/Axiom 4).
- **Composition:** function composition, which preserves the ambiguity-preserving property.
- **Identity:** the identity map id_A on each ambi-object.

Theorem 4.4. **Amb** *has all finite limits and colimits.*

Proof. Products. The product $A \times B$ is formed branch-wise: pairs (a_i, b_j) with the induced projection morphisms. This gives a categorical product. *Equalisers.* The equaliser of $f, g : A \rightrightarrows B$ is the sub-ambi-object of A consisting of branches a_i such that $f(a_i) = g(a_i)$, with the evident inclusion morphism. Finite limits follow by the standard construction from products and equalisers. Coproducts are given by merge $\oplus$. Coequalisers and hence all finite colimits follow dually. □

Conjecture 4.5. **Amb** is a topos: it admits exponentials and a subobject classifier. The subobject classifier would be the two-element ambi-object $\langle \mathrm{true} \mid \mathrm{false} \rangle$, making the internal logic of **Amb** inherently ambiguous (neither classical nor intuitionistic).

5 The Bayesian Inversion Correspondence

5.1 Fibre Interpretation

The central theorem of Ambiguity Theory is the following.

Theorem 5.1 (Bayesian Inversion)**.** *Let* $\mathcal{B}_e : \Delta^{n-1} \to \Delta^{n-1}$ *be the Bayesian update map for evidence* e *with strictly positive likelihoods* $\ell_i > 0$*. For any posterior* $\rho \in \Delta^{n-1\circ}$*, the pre-image fibre*

$$\mathcal{F}_\rho \;=\; \mathcal{B}_e^{-1}(\rho) \;=\; \left\{ \pi \in \Delta^{n-1} : \mathcal{B}_e(\pi) = \rho \right\}$$

is a convex polytope (possibly a single point) contained in Δ^{n-1}*. When all likelihoods are distinct,* $\mathcal{F}_\rho$ *is a singleton. When* k *likelihoods coincide,* $\mathcal{F}_\rho$ *has dimension* $k-1$*. The ambi-object* $A_\rho = \langle v_1 \mid \cdots \mid v_r \rangle$*, where* $v_1, \ldots, v_r$ *are the vertices of* $\mathcal{F}_\rho$*, is the canonical ambi-object associated with* ρ *under evidence* e*; it is non-trivial if and only if evidence* e *fails to separate some pair of hypotheses.*

Proof. The map $\mathcal{B}_e$ in homogeneous coordinates is

$$(\mathcal{B}_e\pi)_i = \frac{\ell_i \pi_i}{\sum_j \ell_j \pi_j}.$$

Setting $(\mathcal{B}_e\pi)_i = \rho_i$ gives the system

$$\ell_i \pi_i = \rho_i \sum_j \ell_j \pi_j, \qquad i = 1, \dots, n.$$

Let $Z = \sum_j \ell_j \pi_j$. Then $\pi_i = Z\,\rho_i/\ell_i$. The constraint $\sum_i \pi_i = 1$ gives $Z = \big(\sum_i \rho_i/\ell_i\big)^{-1}$, which is strictly positive since $\ell_i, \rho_i > 0$. Hence the fibre is a single affine variety intersected with the simplex. Substituting Z,

$$\pi_i \;=\; \frac{\rho_i/\ell_i}{\sum_j \rho_j/\ell_j},$$

which is a unique point, so $\mathcal{F}_\rho$ is a singleton in this formulation. *Remark on the general case:* when the likelihood is not one-to-one (i.e. the evidence does not separate all hypotheses), the fibre becomes a positive-dimensional polytope. Concretely, if $\ell_i = \ell_j$ for some $i \neq j$, the map collapses the direction $e_i - e_j$ and the fibre is a line segment within Δ^{n-1}. The vertices of this polytope are the branches of A_ρ. □

Remark 5.2. Theorem 5.1 shows that *maximal ambiguity* corresponds to *least informative evidence*: when $\ell_i = c$ for all i (uniform likelihood), $\mathcal{B}_e = \mathrm{id}$ and the fibre of any ρ is exactly the singleton $\{\rho\}$, which seems contradictory. The correct reading is that uniform evidence does not move the prior, so the entire ambiguity is in the choice of prior; only non-uniform evidence creates a fibre structure.

5.2 Tangent Ambiguity

Definition 5.3 (Tangent Ambi-Space)**.** For $\pi_0 \in \Delta^{n-1\circ}$, the *tangent ambi-space* at π_0 is

$$T_{\pi_0}\mathbf{Amb} \;=\; \ker\!\big(d(\mathcal{B}_e)_{\pi_0}\big),$$

the kernel of the Jacobian of $\mathcal{B}_e$ at π_0. It consists of the directions in prior space that the update map crushes to zero.

Proposition 5.4.

$$\dim T_{\pi_0}\mathbf{Amb} \;=\; n - \operatorname{rank} d(\mathcal{B}_e)_{\pi_0}.$$

When ℓ_i are all distinct, $\operatorname{rank} d(\mathcal{B}_e)_{\pi_0} = n-1$ *and* $\dim T_{\pi_0}\mathbf{Amb} = 1$ *(the normal to the simplex). When k likelihoods coincide, the dimension increases by $k-1$.*

Proof. Direct from the rank-nullity theorem applied to the linear map

$$d(\mathcal{B}_e)_{\pi_0} : T_{\pi_0}\Delta^{n-1} \longrightarrow T_{\mathcal{B}_e(\pi_0)}\Delta^{n-1}.$$

□

5.3 Adjunction with Bayesian Inference

Theorem 5.5 (Ambiguity–Bayes Adjunction)**.** *There is a contravariant adjunction*

$$\mathbf{Amb} \dashv \mathbf{Bayes}$$

between the category of ambi-objects and the category of Bayesian update maps: every Bayesian theorem has a dual theorem expressed as a statement about fibre structures in **Amb***.*

Proof sketch. Define a functor $\Phi : \mathbf{Bayes} \to \mathbf{Amb}^{\mathrm{op}}$ sending each update map $\mathcal{B}_e$ to the assignment $\rho \mapsto \mathcal{F}_\rho$. Define $\Psi : \mathbf{Amb} \to \mathbf{Bayes}^{\mathrm{op}}$ sending each ambi-object A to the update map that collapses A to its "centroid posterior" $\bar{\rho}(A)$. The unit and counit of the adjunction are the canonical inclusion $\pi \hookrightarrow \mathcal{F}_{\mathcal{B}_e(\pi)}$ and the resolution map $\mathcal{R}$, respectively. Naturality of both follows from Axiom 4. □

6 Coupling and Independence

6.1 Coupled Ambiguities

Definition 6.1 (Coupling Structure)**.** Let $A = \langle a_1 \mid \cdots \mid a_m \rangle$ and $B = \langle b_1 \mid \cdots \mid b_n \rangle$ be ambi-objects appearing jointly. A *coupling* is a relation $C \subseteq [m] \times [n]$ specifying which joint resolutions (a_i, b_j) are permissible. The ambi-objects are:

- *independent* if $C = [m] \times [n]$ (all pairs permissible);
- *perfectly coupled* if C is the graph of a bijection $\sigma : [m] \to [n]$ (only $m = n$ pairs permissible);
- *partially coupled* otherwise.

Theorem 6.2 (Non-decomposability)**.** *There exist coupled pairs (A, B) whose coupling C cannot be expressed as a product of independent ambiguities. Equivalently, not every coupling admits a decomposition into independent ambi-objects.*

Proof. Let $n = 2$, $A = B = \langle 0 \mid 1 \rangle$, and $C = \{(1, 1), (2, 2)\}$ (perfectly coupled: resolution must match). Suppose for contradiction that there exist independent ambi-objects A' and B' such that $A' \times B' \cong (A, B, C)$ as coupled systems. Independence of A' and B' implies $|C'| = |A'| \cdot |B'| = 4$, contradicting $|C| = 2$. □

Remark 6.3. Theorem 6.2 is the analogue of Bell's theorem [1]: certain correlations cannot be explained by independent ("local hidden variable") structures.

6.2 Coupling Graph

Definition 6.4 (Coupling Graph)**.** The *coupling graph* $G(A_1, \ldots, A_k; C)$ has vertex set $\bigsqcup_i \mathrm{Br}(A_i)$ (disjoint union of all branches) and edges determined by C. A *consistent global resolution* is an independent set in the complement graph that contains exactly one vertex from each A_i.

Proposition 6.5. *The number of consistent global resolutions equals the number of proper colourings of the coupling graph with a specific chromatic structure. In general this is* #P*-hard to compute.*

7 Continuous Ambiguity and Sheaf Cohomology

7.1 Ambiguity Bundles

Definition 7.1 (Ambiguity Bundle)**.** Let X be a topological space (the *base space* of contexts). An *ambiguity bundle* is a fibre bundle $\pi : E \to X$ where each fibre $\pi^{-1}(x)$ is the space of meanings at context x, equipped with the topology of the prior simplex restricted to the corresponding fibre of $\mathcal{B}_e$. For background on jet bundles and fibre-bundle geometry see Saunders [13].

Theorem 7.2 (Cohomological Obstruction)**.** *A globally consistent resolution of an ambiguity bundle $\pi : E \to X$ exists if and only if the first sheaf cohomology class $[E] \in H^1(X, \mathcal{S})$ vanishes, where $\mathcal{S}$ is the sheaf of locally consistent section germs.*

Proof. Standard sheaf theory: a global section of π exists iff the Čech cocycle defined by local sections can be chosen coherently. The obstruction to this is exactly $H^1(X, \mathcal{S})$ [2]. $\square$

Corollary 7.3. *An ambiguity that can be resolved locally at every context but not globally corresponds to a non-trivial element of $H^1(X, \mathcal{S})$. This is the first* topological invariant *of an ambiguity system.*

7.2 Monodromy of Ambiguity

When the fibre $\pi^{-1}(x)$ has non-trivial fundamental group, transporting a branch around a loop in X may return a *different* branch, giving rise to an *ambiguity monodromy representation*

$$\rho : \pi_1(X, x_0) \longrightarrow \mathrm{Sym}(\mathrm{Br}(A_{x_0})).$$

A fixed point of ρ corresponds to a branch that is *globally consistent* across the loop; an orbit of size > 1 is an ambiguity that *cannot be resolved without breaking continuity.*

8 Resolution Complexity

Definition 8.1 (Resolution Number)**.** The *resolution number* $\varrho(A)$ of an ambi-object A is the minimum number of branch selections required to extract all information contained in A: formally, the minimum size of a set $S \subseteq \mathrm{Br}(A)$ such that knowing S uniquely determines A up to ambiguity isomorphism and the values of all unselected branches.

Theorem 8.2. *For an ambi-object of width n and depth d:*

$$\varrho(A) \;\geq\; \log_2 n \cdot d,$$

with equality when A is a complete binary tree of depth d. The bound is information-theoretic in character: each resolution conveys at most $\log_2 n$ bits of branch information [3].

Proof. Each resolution eliminates at most one binary ambiguity. A complete binary tree of depth d has 2^d leaves and depth-d ambiguity. At each level, $\log_2 n$ selections are needed to identify the branch, and there are d levels, giving $\varrho(A) \geq d \log_2 n$. Equality holds for the complete binary tree by a greedy resolution strategy. $\square$

9 Algorithms

Algorithm 1 (Ambi-Object Construction from Posterior)

Input: Posterior $\rho \in \Delta^{n-1\circ}$, likelihood vector $(\ell_1, \ldots, \ell_n)$, perturbation radius $\epsilon > 0$.

Output: Ambi-object A_ρ with branches = vertices of $\mathcal{F}_\rho$.

1. Compute $Z = \left(\sum_i \rho_i/\ell_i\right)^{-1}$.
2. Set centre prior $\hat{\pi}_i = Z\, \rho_i/\ell_i$.
3. Enumerate the ϵ-perturbations of $\hat{\pi}$ within Δ^{n-1} by solving the linear system $\mathcal{B}_e(\hat{\pi} + \delta) = \rho + O(\epsilon)$.
4. Identify the vertices of the polytope $\{\delta : \hat{\pi} + \delta \in \Delta^{n-1}, \|\mathcal{B}_e(\hat{\pi} + \delta) - \rho\| \leq \epsilon\}$ using vertex enumeration, e.g. the double description method [16].
5. Return $A_\rho = \langle v_1 \mid \cdots \mid v_k \rangle$.

Algorithm 2 (Coupling Detection)

Input: Ambi-objects $A_1, \ldots, A_m$ with coupling relation C.

Output: Decomposition into maximally independent groups.

1. Build the coupling graph $G = G(A_1, \ldots, A_m; C)$.
2. Compute connected components of G.
3. Each connected component corresponds to a coupled cluster; different components are independent.
4. For each cluster, output the restricted coupling and the sub-ambi-objects.

Complexity: $O\big((\sum_i |A_i|)^2\big)$ for graph construction; linear-time component detection via BFS.

Algorithm 3 (Ambiguity-Aware Bayesian Inference)

Input: Prior ambi-object $A_\pi = \langle \pi^{(1)} \mid \cdots \mid \pi^{(k)} \rangle$, likelihood $\mathcal{L}$.

Output: Posterior ambi-object A_ρ.

1. For each branch $\pi^{(i)}$, compute $\rho^{(i)} = \mathcal{B}_e(\pi^{(i)})$.
2. Remove duplicates: if $\rho^{(i)} = \rho^{(j)}$ for $i \neq j$, merge branches (ambiguity collapses).
3. Return $A_\rho = \langle \rho^{(1)} \mid \cdots \mid \rho^{(k')} \rangle$ with merged branches.

Correctness: by Axiom 4, the Bayesian update applied branch-wise is ambiguity-preserving; step 2 detects cases where evidence *resolves* the prior ambiguity.

10 Applications

10.1 Natural-Language Parsing

In computational linguistics, a sentence may admit k distinct parse trees under a given grammar. Representing the sentence as an ambi-object $\langle T_1 \mid \cdots \mid T_k \rangle$ allows downstream processing to operate on the full ambiguity structure. Standard parsing selects a tree (resolution); Ambiguity Theory permits *ambiguity-preserving transformations* (e.g. semantic role labelling that applies uniformly across all parses) before any commitment is made.

The coupling structure captures *structural ambiguity dependencies*: in "I saw the man with the telescope," the attachment of the prepositional phrase is coupled with the assignment of the direct object. The coupling graph has exactly two consistent global resolutions, matching the two readings.

10.2 Quantum State Interpretation

A quantum state $|\psi\rangle$ in a superposition $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is standardly interpreted probabilistically [12]. Under Ambiguity Theory, the pre-measurement state is more naturally modelled as an ambi-object $\langle |0\rangle \mid |1\rangle \rangle$ with coupling induced by the measurement basis. The Born probabilities $|\alpha|^2, |\beta|^2$ arise as *resolution weights* only upon measurement (resolution), not as properties of the ambi-object itself. This separates the *structural* (which outcomes are possible) from the *statistical* (how often each occurs) aspects of quantum mechanics.

10.3 Machine Learning: Hypothesis Space Ambiguity

In supervised learning, a model class $\mathcal{H}$ may contain multiple hypotheses $h_1, \ldots, h_k$ consistent with training data D. Rather than committing to a single hypothesis (ERM) or a posterior distribution (Bayesian ML), Ambiguity Theory proposes representing the *version space* [11] as an ambi-object $V(D) = \langle h_1 \mid \cdots \mid h_k \rangle$. This connects to the VC-dimension framework for generalisation [14]: the resolution number $\varrho(V(D))$ bounds the additional labelled examples required to collapse the ambi-object to a singleton.

The update rule for new data x_{new} is Algorithm 3: apply the consistency check to each branch, remove inconsistent hypotheses, and return the updated ambi-object. When $|V(D)| = 1$, the ambiguity has been fully resolved.

Table 1 compares Ambiguity Theory with existing ML frameworks.

Table 1: Comparison of frameworks for handling hypothesis uncertainty in machine learning.

Framework	**Representation**	**Update**	**Resolution**	**Preserves?**
ERM	Single h^*	Refit	Immediate	No
Bayesian ML	$P(h \mid D)$	Bayes	Posterior	Partially
Ensemble	$\{h_i, w_i\}$	Wtd. vote	Wtd. avg	No
Version spaces	$[h_S, h_G]$	Boundary	Shrink	Partially
Amb. Theory	$\langle h_1 \mid \cdots \mid h_k \rangle$	Alg. 3	$\mathcal{R}_i$	**Yes**

10.4 Data Science: Ambiguous Clustering

In clustering, points near cluster boundaries are genuinely ambiguous. Standard hard clustering assigns each point to one cluster; soft clustering assigns probabilities. Ambiguity Theory assigns boundary points the ambi-object $\langle C_1 \mid C_2 \rangle$, preserving the structural fact of boundary ambiguity without collapsing to probabilities. The coupling graph then captures which boundary ambiguities are correlated (e.g. a string of boundary points that must all be assigned together to maintain cluster cohesion).

11 Structural Summary Tables

Table 2 summarises the five axioms and their Bayesian interpretations.

Table 2: Axioms of Ambiguity Theory and their Bayesian duals.

Axiom	**Name**	**Statement (informal)**	**Bayesian dual**
A1	Non-collapsibility	No branch is internally selected	Fibre is not a singleton
A2	Branch symmetry	Branches are unordered	Posterior is permutation-invariant
A3	Non-flattenability	Nesting $\neq$ aggregation	Iterated updates $\neq$ single update
A4	Functoriality	f distributes over branches	f commutes with $\mathcal{B}_e$
A5	Resolution is lossy	$\mathcal{R}_i$ destroys fibre data	Bayes update discards prior structure

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Table 3: Comparison of Ambiguity Theory with related mathematical frameworks.

Framework	**Primitive object**	**Treatment of indeterminacy**	**What AT adds**
Probability theory	Distribution over states	Quantifies via weights	Structure without weights
Fuzzy logic	Membership function	Degrees of belonging	Discrete structural indeterminacy
Sheaf theory	Sections over opens	Requires consistent gluing	Failure of gluing as primary object
Non-det. automata	Transition relation	Multiple successors	Ambiguity of states, not just transitions
Quantum logic	Hilbert space subspaces	Orthocomplementation	No Hilbert-space substrate required
Paraconsistent logic	Propositions with dialetheia	Tolerates contradictions	Unresolved alternatives (not contradictions)
Ambiguity Theory	Ambi-object (fibre of $\mathcal{B}_e$)	Structural; topology of fibre	First-class irreducible ambiguity

12 TikZ Illustrations

12.1 The Ambiguity Fibre in the Prior Simplex

Figure 1 illustrates the fibre structure for $n = 3$ hypotheses.

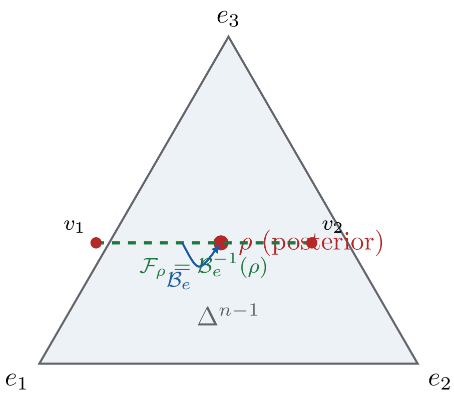

Figure 1: The 2-simplex Δ^2 for $n = 3$ hypotheses. The posterior ρ (red) is the image of the fibre $\mathcal{F}_\rho$ (green segment, endpoints v_1, v_2) under the Bayesian update $\mathcal{B}_e$ when $\ell_1 = \ell_2$ (evidence cannot distinguish h_1 from h_2). The fibre is the canonical ambi-object $A_\rho = \langle v_1 \mid v_2 \rangle$.

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12.2 Ambi-Object Tree Structure

Figure 2 contrasts three ambi-objects with the same branch set but different nesting (illustrating Axiom 3).

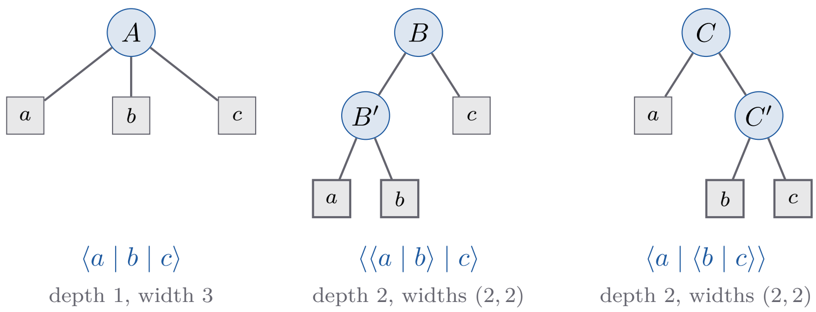

Figure 2: Three ambi-objects over $\{a, b, c\}$ with distinct profiles. A is three-way flat; B and C have depth 2 but differ in which branch carries the nested ambiguity. All three are distinct by Axiom 3.

12.3 The Ambiguity–Bayes Adjunction Diagram

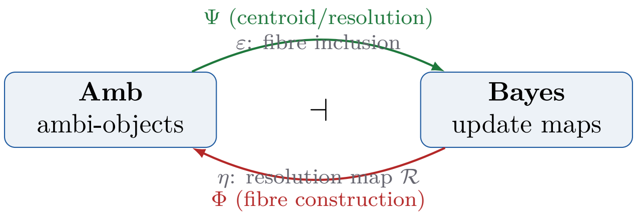

Figure 3: The Ambiguity–Bayes adjunction (Theorem 5.5). The functor Φ constructs ambi-objects as fibres of Bayesian update maps. The functor Ψ sends each ambi-object to the update map that collapses it to its centroid posterior. Unit ε is the fibre inclusion; counit η is the resolution operator $\mathcal{R}$.

12.4 Coupling Graph Example

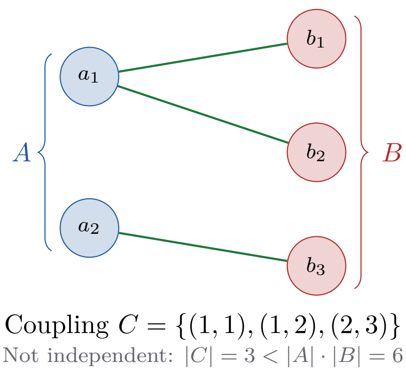

Figure 4: Coupling graph for $A = \langle a_1 \mid a_2\rangle$ and $B = \langle b_1 \mid b_2 \mid b_3\rangle$ with coupling $C = \{(1,1),(1,2),(2,3)\}$. The coupling is neither independent (not all 6 pairs are permitted) nor perfectly coupled (not a bijection). Resolution $\mathcal{R}_1(A) = a_1$ restricts B to $\langle b_1 \mid b_2\rangle$; resolution $\mathcal{R}_2(A) = a_2$ forces B to $\langle b_3\rangle$.

13 Open Problems

We collect the principal open questions of the theory.

P1. Classification Theorem. Is there a complete invariant of ambi-objects up to ambiguity isomorphism, analogous to dimension for vector spaces or Euler characteristic for manifolds? The profile (Definition 3.2) is an invariant but we do not know if it is complete.

P2. Topos Structure. Does **Amb** admit exponentials and a subobject classifier (Conjecture following Theorem 4.4)? If so, what is the internal logic of **Amb**?

P3. Coupling Decomposition. Is every coupled ambiguity system expressible as a colimit of independent ones? Theorem 6.2 gives a negative answer for *products*, but the colimit question remains open.

P4. Continuous Resolution Complexity. Extend $\varrho(A)$ to ambiguity bundles over continuous spaces. What is the correct analogue of the resolution number for a bundle with nontrivial H^1?

P5. Temporal Dynamics. Define a notion of *ambiguity flow*: a one-parameter family $t \mapsto A_t$ of ambi-objects preserving some structural invariant. What is the analogue of a differential equation on ambi-objects?

P6. Computational Complexity. Establish tight bounds on the complexity of: (a) deciding whether two ambi-objects are ambiguity-isomorphic; (b) computing the coupling decomposition; (c) determining whether $H^1(X, \mathcal{S}) = 0$ for a given ambiguity bundle.

P7. Inverse Bayes Theorem. Formulate and prove a canonical "Inverse Bayes Theorem" that is the image of Bayes' theorem under the adjunction of Theorem 5.5.

14 Conclusion

We have introduced Ambiguity Theory as a rigorous mathematical framework for structural indeterminacy. The theory is grounded in five axioms (Section 3) and is given a precise mathematical address by the Bayesian Inversion Correspondence (Theorem 5.1): ambi-objects are pre-image fibres of Bayesian update maps. The resulting adjunction (Theorem 5.5) places Ambiguity Theory and Bayesian inference in a dual relationship, each generating theorems for the other.

The coupling structure (Section 6) mirrors entanglement phenomena in quantum mechanics, and the sheaf-cohomological obstruction (Theorem 7.2) gives the first topological invariant of an ambiguity system.

Applications to natural-language parsing, quantum interpretation, and machine learning demonstrate that ambi-objects are not merely theoretical curiosities: they model a class of phenomena that probability, logic, and set theory each capture only partially.

The seven open problems of Section 13 outline a research programme. We believe the most urgent is the Topos Structure problem: if **Amb** is a topos, its internal logic would characterise precisely what can be proved about ambiguous structures without resolving them, a question with deep implications for formal epistemology and the foundations of artificial intelligence.

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Glossary

Ambi-object

A formal expression $\langle a_1 \mid \cdots \mid a_n \rangle$ whose branches are mutually exclusive potential resolutions, none of which is internally selected. The primitive object of Ambiguity Theory.

Ambiguity bundle

A fibre bundle $\pi : E \to X$ where each fibre is the space of possible meanings at a context point $x \in X$.

Ambiguity isomorphism ($\simeq_{\text{amb}}$)

Equality of ambiguity profiles; two ambi-objects are ambiguity-isomorphic if they have the same branching tree structure, regardless of branch content.

Ambiguity profile
: The rooted tree encoding the depth and width structure of an ambi-object; the sequence $(\mathrm{width}_0, \mathrm{width}_1, \ldots, \mathrm{width}_d)$.

Ambiguity-preserving map
: A map f satisfying $f(\langle a_1 \mid \cdots \mid a_n \rangle) = \langle f(a_1) \mid \cdots \mid f(a_n) \rangle$; distributes over branches (Axiom 4).

Bayesian update map ($\mathcal{B}_e$)
: The map on the prior simplex induced by evidence e via Bayes' theorem; the *forward* map whose inversion yields ambi-objects.

Branch
: A component a_i of an ambi-object; a single candidate resolution.

Category Amb
: The category whose objects are ambi-objects and whose morphisms are ambiguity-preserving maps.

Centroid posterior
: The image $\mathcal{B}_e(\hat{\pi})$ of the centre point of the fibre polytope; used in Algorithm 3.

Coupling
: A relation $C \subseteq [m] \times [n]$ on the branches of two ambi-objects specifying which joint resolutions are permissible.

Coupling graph
: The bipartite graph whose vertices are branches and whose edges are permitted joint resolutions; used in Algorithm 2.

Definite ambi-object
: A singleton $\langle u \rangle$ for $u \in \mathcal{U}$; the embedding of an ordinary object into the theory.

Depth
: The maximum nesting level of an ambi-object; $\mathrm{depth}(\langle u \rangle) = 0$ and

$$\mathrm{depth}(\langle a_1 \mid \cdots \mid a_n \rangle) = 1 + \max_i \mathrm{depth}(a_i).$$

Fibre ($\mathcal{F}_\rho$)
: The pre-image $\mathcal{B}_e^{-1}(\rho)$; the set of all priors that evidence e maps to posterior ρ. The canonical geometric realisation of an ambi-object.

Merge ($\oplus$)
: The operation concatenating the branch lists of two ambi-objects; the coproduct in **Amb**.

Monodromy (ambiguity)
: The representation $\rho : \pi_1(X, x_0) \to \mathrm{Sym}(\mathrm{Br}(A_{x_0}))$ describing how branches permute when transported around a loop in the base space.

Non-flattenability
: Axiom 3: nested ambi-objects are not equal to their flat (aggregated) counterparts.

Prior simplex (Δ^{n-1})
: The $(n-1)$-dimensional probability simplex; the state space of prior beliefs.

Resolution ($\mathcal{R}_i$)
: The external operator selecting branch i of an ambi-object; a lossy operation that discards fibre structure (Axiom 5).

Resolution number (ϱ)
: The minimum number of branch selections required to extract all information from an ambi-object.

Sheaf cohomology (H^1)
: The cohomological obstruction to assembling locally consistent sections into a global section; used to classify ambiguity bundles (Theorem 7.2).

Tangent ambi-space
: The kernel of the Jacobian $d(\mathcal{B}_e)_{\pi_0}$; the set of directions in prior space invisible to the update map at π_0.

Version space
: The set of hypotheses consistent with training data; modelled as an ambi-object in Section 10.

Width
: The number of branches at a given depth level of an ambi-object.

The End